

Supplement: The Bridge Theorem for 1/f Frame-Edge Signatures

Scale Invariance, Log-Uniform Timescales, and Dimensional Ladder Transitions in the Lambda / TuFT / NSAF Framework

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Status: Timestamped independent work in progress.

Relation to TUFT: TUFT refers here to Jennifer “Jenny” Lorraine Nielsen’s *Topological Unified Field Theory on the Complex Hopf Fibration*. Lambda supplies the residue-accounting and frame-transition language used in this supplement. NSAF supplies the attractor / recurrence / scale-cascade interpretation.

Non-claim statement: This supplement does not claim ownership of established results in stochastic processes, spectral analysis, harmonic analysis, probability theory, mathematical physics, topology, Hopf fibrations, K-theory, or gauge-bundle theory. It records an independent framework-level synthesis and a bridge theorem explaining when 1/f-type spectra follow from scale-invariant frame transitions.

1 Purpose

The prior supplement established the conditional mathematical fact

$$g(\tau) \propto \frac{1}{\tau} \implies S(f) \propto \frac{1}{f} \quad (1)$$

for a superposition of Lorentzian relaxation modes over an intermediate frequency band.

That derivation is valid, but it leaves one missing bridge. The derivation proves that a log-uniform hierarchy of relaxation or integration timescales produces a 1/f spectrum. It does not by itself prove that a Lambda / TuFT dimensional ladder transition generates such a hierarchy.

This supplement supplies that bridge.

The central claim is:

A genuine dimensional ladder transition has no privileged intermediate scale. Therefore the unresolved middle of the transition carries the multiplicative scale measure $d\tau/\tau$, not the additive measure $d\tau$. Consequently, where the unresolved modes contribute Lorentzian relaxation spectra, the aggregate frame-edge residue has 1/f form in the intermediate band.

This gives the full theorem chain:

$$\text{genuine dimensional ladder transition} \Rightarrow \text{scale-invariant unresolved middle} \Rightarrow g(\tau) \propto \frac{1}{\tau} \Rightarrow S(f) \propto \frac{1}{f}. \quad (2)$$

2 Conceptual Setting

The Lambda / TuFT / NSAF synthesis distinguishes two representational profiles:

- P_C : a curved, global, continuous, spectral, bundle-theoretic, or higher-dimensional paradigm.
- P_L : a linear, local, discrete, sampled, arithmetic, observed, or lower-dimensional paradigm.

A frame transition

$$T : P_C \rightarrow P_L \quad (3)$$

attempts to render some higher global structure inside a lower observational or formal frame.

Examples include a probability law represented by finite samples; a continuous signal represented by discrete measurements; a global spectral measure represented by local data; a nontrivial bundle represented by local trivializations; and a curved dynamical field represented by discrete events or observations.

Probability plays a special role in this setting. It is not merely another example. Probability is the natural bridge object between local discrete samples and global continuum structure:

$$\text{discrete samples} \longrightarrow \text{empirical distribution} \longrightarrow \text{probability measure / spectral law.} \quad (4)$$

Thus probability mediates between P_L and P_C . The 1/f question asks what spectral signature appears when that mediating layer contains unresolved structure at every multiplicative scale.

3 Definitions

Definition 1 (Frame Transition). *A frame transition is a representation map*

$$T : P_C \rightarrow P_L \quad (5)$$

from a higher, curved, continuous, global, or spectral paradigm into a lower, linear, discrete, local, or sampled paradigm.

The transition is not required to be invertible. In Lambda language, failure of faithful round-trip recovery is exactly where residue may appear.

Definition 2 (Dimensional Ladder Transition). *A dimensional ladder transition is a frame transition between adjacent representational rungs, where the upper rung has more global, continuous, spectral, or bundle-level information than the lower rung can directly encode.*

A dimensional ladder transition has two boundary scales:

- $\tau_{\min}$: *the lower observational, sampling, or local resolution scale;*
- $\tau_{\max}$: *the upper coherence, integration, or global-reconstruction scale.*

The unresolved middle is the open scale interval

$$(\tau_{\min}, \tau_{\max}). \quad (6)$$

Definition 3 (Genuine Dimensional Ladder Transition). *A dimensional ladder transition is genuine when no additional structure in the transition selects a preferred intermediate scale*

$$\tau_0 \in (\tau_{\min}, \tau_{\max}). \quad (7)$$

Equivalently, the transition distinguishes the endpoint roles of local sampling and global coherence, but it does not privilege one intermediate octave, decade, or scale band over another.

This definition captures the intended Lambda / TuFT case: the transition crosses a structural seam between different representational rungs rather than merely rescaling within a single frame.

Definition 4 (Scale-Invariant Unresolved Middle). *The unresolved middle of a transition is scale-invariant when its description remains invariant under multiplicative rescaling*

$$\tau \mapsto a\tau \quad (8)$$

for admissible $a > 0$ as long as the rescaled interval remains inside the cutoff band.

Scale invariance means that the natural measure of unresolved scale-depth depends on ratios, not absolute differences:

$$\frac{\tau_2}{\tau_1}, \quad \text{not} \quad \tau_2 - \tau_1. \quad (9)$$

4 Bridge Lemma: Scale Invariance Implies Log-Uniformity

Lemma 1 (Haar Measure on the Positive Scale Group). *Let μ be a measure on positive timescales $\tau > 0$ satisfying multiplicative scale invariance:*

$$\mu(aE) = \mu(E) \quad (10)$$

for admissible measurable sets $E \subset (0, \infty)$ and scale factors $a > 0$. Then, up to a constant factor,

$$d\mu(\tau) = \frac{d\tau}{\tau}. \quad (11)$$

Equivalently, the density of scales is

$$g(\tau) = \frac{c}{\tau}. \quad (12)$$

Proof. Set

$$x = \log \tau. \quad (13)$$

Multiplicative rescaling in τ becomes additive translation in x :

$$\tau \mapsto a\tau \iff x \mapsto x + \log a. \quad (14)$$

A measure invariant under all such multiplicative rescalings becomes a translation-invariant measure on the additive line of log-scales. The translation-invariant measure on this line is, up to constant multiple,

$$dx. \quad (15)$$

Since

$$dx = d(\log \tau) = \frac{d\tau}{\tau}, \quad (16)$$

the corresponding scale measure is

$$d\mu(\tau) = c \frac{d\tau}{\tau}. \quad (17)$$

Therefore the density with respect to $d\tau$ is

$$g(\tau) = \frac{c}{\tau}. \quad (18)$$

□

Interpretation. If a transition has no preferred intermediate scale, the correct neutral measure is not uniform in τ . It is uniform in $\log \tau$. Uniformity per octave or per decade is exactly the density needed for 1/f noise.

5 Strong Bridge Theorem: Dimensional Ladder Transitions Are Scale-Invariant

Theorem 1 (Scale Invariance of Genuine Dimensional Ladder Transitions). *Every genuine dimensional ladder transition has a scale-invariant unresolved middle.*

Formally, if

$$T : P_C \rightarrow P_L \quad (19)$$

is a genuine dimensional ladder transition with unresolved scale interval

$$(\tau_{\min}, \tau_{\max}), \quad (20)$$

then the admissible description of its unresolved middle is invariant under multiplicative rescaling

$$\tau \mapsto a\tau \quad (21)$$

within that interval. Consequently, the natural density of unresolved integration or relaxation depths is

$$g(\tau) = \frac{c}{\tau}. \quad (22)$$

Proof. A dimensional ladder transition connects two different representational rungs. The lower rung P_L supplies the local observational cutoff $\tau_{\min}$. The upper rung P_C supplies the global coherence cutoff $\tau_{\max}$. Between these endpoints lies the unresolved middle:

$$(\tau_{\min}, \tau_{\max}). \quad (23)$$

Because the transition is genuine, no additional datum selects a privileged intermediate scale τ_0 . If such a privileged τ_0 were selected, the transition would no longer be a pure ladder transition. It would include an additional scale-setting structure, section, cutoff, clock, damping constant, renormalization anchor, or chosen intermediate trivialization.

Therefore, the unresolved middle contains no invariant additive unit of scale. The only invariant comparisons available inside the middle are ratios:

$$\frac{\tau_2}{\tau_1}. \quad (24)$$

Ratio comparison is multiplicative. Hence the admissible transformations of the unresolved middle are rescalings

$$\tau \mapsto a\tau. \quad (25)$$

By Lemma 4.1, the unique neutral measure compatible with this multiplicative invariance is

$$d\mu(\tau) = c \frac{d\tau}{\tau}. \quad (26)$$

Thus every genuine dimensional ladder transition has a log-uniform unresolved scale measure:

$$g(\tau) = \frac{c}{\tau}. \quad (27)$$

□

Clarification. The theorem does not say that every arbitrary sampling process produces 1/f noise. It says that a genuine dimensional ladder transition, as defined above, has a scale-invariant unresolved middle. If the physical, computational, or observational realization of that middle decomposes into relaxation or integration modes, then those modes are naturally distributed log-uniformly unless a privileged scale is added.

6 Spectral Consequence: 1/f from Scale-Invariant Frame Edges

Proposition 1 (Lorentzian Superposition Produces 1/f Noise). *Suppose each unresolved mode with characteristic time τ contributes a Lorentzian spectrum*

$$S_\tau(f) = \frac{2\sigma^2\tau}{1 + (2\pi f\tau)^2}. \quad (28)$$

Suppose also that the unresolved modes are distributed log-uniformly over $[\tau_{\min}, \tau_{\max}]$, so that

$$g(\tau) = \frac{c}{\tau}. \quad (29)$$

Then the aggregate spectrum satisfies

$$S(f) \propto \frac{1}{f} \quad (30)$$

in the intermediate frequency regime.

Proof. The aggregate spectrum is

$$S(f) = \int_{\tau_{\min}}^{\tau_{\max}} \frac{2\sigma^2\tau}{1 + (2\pi f\tau)^2} \frac{c}{\tau} d\tau. \quad (31)$$

Therefore

$$S(f) = 2\sigma^2 c \int_{\tau_{\min}}^{\tau_{\max}} \frac{1}{1 + (2\pi f\tau)^2} d\tau. \quad (32)$$

Let

$$u = 2\pi f\tau. \quad (33)$$

Then

$$d\tau = \frac{du}{2\pi f}, \quad (34)$$

and

$$S(f) = \frac{\sigma^2 c}{\pi f} [\arctan(u)]_{2\pi f\tau_{\min}}^{2\pi f\tau_{\max}}. \quad (35)$$

Hence

$$S(f) = \frac{\sigma^2 c}{\pi f} [\arctan(2\pi f\tau_{\max}) - \arctan(2\pi f\tau_{\min})]. \quad (36)$$

In the intermediate frequency regime

$$2\pi f\tau_{\min} \ll 1 \ll 2\pi f\tau_{\max}, \quad (37)$$

we have

$$\arctan(2\pi f\tau_{\min}) \approx 0, \quad (38)$$

and

$$\arctan(2\pi f\tau_{\max}) \approx \frac{\pi}{2}. \quad (39)$$

Thus

$$S(f) \approx \frac{\sigma^2 c}{2f}. \quad (40)$$

Therefore

$$S(f) \propto \frac{1}{f}. \quad (41)$$

□

7 Full Bridge Theorem

Theorem 2 (1/f Spectral Residue at Genuine Dimensional Ladder Transitions). *Let*

$$T : P_C \rightarrow P_L \quad (42)$$

be a genuine dimensional ladder transition. Suppose the unresolved middle of the transition is realized by independent or weakly coupled relaxation / integration modes, each with Lorentzian spectral contribution

$$S_\tau(f) = \frac{2\sigma^2\tau}{1 + (2\pi f\tau)^2}. \quad (43)$$

Then the aggregate unresolved spectrum satisfies

$$S(f) \sim \frac{1}{f} \quad (44)$$

in the intermediate frequency regime

$$1/\tau_{\max} \ll f \ll 1/\tau_{\min} \quad (45)$$

up to the conventional factor 2π .

Proof. By Theorem 5.1, every genuine dimensional ladder transition has a scale-invariant unresolved middle. By Lemma 4.1, scale invariance implies the log-uniform timescale density

$$g(\tau) = \frac{c}{\tau}. \quad (46)$$

By Proposition 6.1, a log-uniform superposition of Lorentzian relaxation modes produces

$$S(f) \propto \frac{1}{f} \quad (47)$$

in the intermediate frequency band.

Therefore a genuine dimensional ladder transition realized through Lorentzian relaxation or integration modes produces 1/f spectral residue in the intermediate regime. $\square$

8 Relation to TUFT

TUFT, as used here, refers to Nielsen's Topological Unified Field Theory on the complex Hopf fibration. The TUFT canon presents unification through nontrivial bundle topology rooted in the complex Hopf fibration, including the finite shell

$$S^1 \rightarrow S^9 \rightarrow CP^4, \quad (48)$$

and the universal complex Hopf fibration

$$S^1 \rightarrow S^\infty \rightarrow CP^\infty. \quad (49)$$

In the present supplement, Lambda does not redefine TUFT. Rather:

- TUFT supplies the topological / bundle context in which nontrivial fiber-base structure matters.

- Lambda supplies the residue-accounting language for frame transitions.
- NSAF supplies the attractor / cascade language for how unresolved structure can persist across scales.

The bridge theorem uses this shared architecture without claiming that the 1/f theorem has already been proved in the TUFT source. The supplement adds a Lambda / TUFT-compatible theorem:

If a nontrivial dimensional ladder transition has no preferred intermediate scale, then the unresolved middle carries log-uniform scale measure; when realized by Lorentzian relaxation modes, it produces 1/f spectral residue.

This is a natural extension of the TUFT idea that product-space or trivialized descriptions lose the very topological information that makes the bundle physically meaningful. In Lambda language, the lost or unresolved information appears as typed residue. In the spectral case considered here, that residue has 1/f form when the transition remains scale-invariant.

9 Probability as the Bridge Object

Probability should be treated as a central bridge rather than a peripheral example.

A single outcome belongs to P_L :

$$\text{heads, tails, 1, 2, } \dots \quad (50)$$

A probability law belongs to P_C :

$$([0, 1], \mathcal{B}, \lambda), \quad (51)$$

or more generally a measure space

$$(\Omega, \mathcal{F}, \mathbb{P}). \quad (52)$$

The empirical distribution is the bridge between them:

$$\frac{1}{N} \sum_{n=1}^N \delta_{X_n} \longrightarrow \mathbb{P}. \quad (53)$$

When convergence occurs through independent identically distributed samples with a single natural scale, the raw sequence may exhibit white-noise behavior. Thus ordinary sampling alone does not imply 1/f.

When the bridge contains unresolved hierarchy across scales, the empirical reconstruction does not proceed through one flat sampling depth. It proceeds through nested integration depths, memory windows, relaxation constants, or coarse-graining levels. If those levels have no privileged scale, Theorem 7.1 applies.

Thus probability provides the general bridge, while 1/f identifies a special bridge condition:

$$\text{probability bridge} + \text{scale-invariant unresolved hierarchy} \Rightarrow \text{1/f spectral residue}. \quad (54)$$

10 Status of Applications

The theorem applies directly where the following conditions can be justified:

1. A frame transition $P_C \rightarrow P_L$ is present.
2. The transition is a genuine dimensional ladder transition with no privileged intermediate scale.
3. The unresolved middle is represented by relaxation or integration modes.
4. The modes are independent or weakly coupled enough for spectral superposition.
5. The observed frequency band lies between the inverse cutoffs.

Applications to neural dynamics, heart-rate variability, turbulence, financial volatility, quantum measurement, and other scale-rich phenomena should be classified as interpretive applications unless these conditions are explicitly verified in the target system.

This status separation preserves the theorem while allowing the broader research program to remain generative.

11 Final Synthesis

The bridge theorem resolves the gap.

The valid mathematical chain is

$$\text{no privileged intermediate scale} \Rightarrow \text{multiplicative scale invariance} \quad (55)$$

$$\Rightarrow \text{log-uniform timescale density} \quad (56)$$

$$\Rightarrow 1/f \text{ spectral residue.} \quad (57)$$

For genuine dimensional ladder transitions, the absence of a privileged intermediate scale follows from the nature of the transition itself: the transition connects different representational rungs and supplies endpoint cutoffs, but no internal scale anchor.

Therefore, within the Lambda / TuFT / NSAF framework:

1/f noise is the spectral residue of a genuine dimensional ladder transition when the unresolved scale-invariant middle is realized through relaxation or integration modes.

This is the stronger bridge. It does not claim that arbitrary sampling produces 1/f. It claims that genuine scale-invariant frame transitions do, and it proves why.